Witch Hunt- Monster Shooter 2D game

SE4031- Game Document

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Date of submission – 03/29/2026

1 Concept

Witch Hunt is a fast-paced, top-down 2D action game where players take control of a witch navigating through various treacherous environments. The core gameplay revolves around fluid movement and precision aiming. As the witch, the player must fend off supernatural entities Bats, Ghosts, and the formidable Pumpkin Enemies using magical projectiles.

The game balances high-intensity combat with strategic resource management, requiring players to utilize "Speed Rings" for mobility and "Health Potions" for survival. With three distinct levels ranging from jungle skirmishes to timed survival challenges, *Witch Hunt* offers a progressive difficulty curve designed to test the player's reflexes and tactical positioning.

Figure 1 : Main menu

2 Core Features & Implementation

The game is built in Unity, utilizing a 2D top-down perspective with a focus on physics-based collisions and dynamic UI feedback.

2.1 Mechanics

- **Movement & Aiming:** The player uses a classic WASD configuration for movement. The shooting mechanic is decoupled from movement, allowing the player to aim 360 degrees toward the mouse cursor location.

- **Health System:** The player starts with 3 life points. This is represented in the UI by three heart/health indicators. Taking damage from enemy projectiles or collisions depletes these points; while picking up health potions restores them.

2.2 Enemy Behavior and Variety

Bats: Aggressive "suicide" units that idle until the player is within range, then charge directly at the witch. They are fragile (1-hit kill) but highly dangerous.

Figure 2 : Bat

Ghosts: Ranged attackers that maintain a slow pace toward the player while firing projectiles. They feature a higher health pool than bats.

Figure 3: Ghost Enemy

Pumpkin Enemies: Mini-bosses that fire multiple projectiles simultaneously, creating "bullet-hell" scenarios.

Figure 4: Pumpkin mini boss

Hit Feedback

Both Ghosts and Pumpkins utilize a dynamic "On-Hit" health bar that appears only when damage is dealt and fades out shortly after, keeping the screen clean during exploration.

2.3 Environmental Interactables

- **Speed Rings:** Inspired by classic arcade mechanics, passing through these rings provides a temporary velocity boost. This is essential for dodging large groups of enemies or clearing the "Escape Dungeon" level.
- **Obstacles:** Trees and walls act as physical barriers that both the player and enemies must navigate around, creating natural "choke points" for combat.

Figure 5: Health potion

Figure 6: Health potion Taken

Figure 7: Speed Rings

3 Level Design

Level 1: The Jungle Hunt

This introductory level serves as a "Search and Destroy" mission set within a dense, overgrown forest. The primary goal is to clear the map of all hidden threats to proceed.

- **Environment & Hazards:** The map is cluttered with tree obstacles that serve as natural cover. However, these trees also block the player's line of sight, allowing **Bats** to ambush the witch from behind foliage.
- **Gameplay Mechanics:** Players are introduced to the **Speed Rings** scattered throughout the jungle. Utilizing these is key to quickly closing the distance between enemy clusters or escaping a coordinated strike from **Ghosts**.
- **Tactical Element:** Because Ghosts can shoot through some gaps where the player cannot easily move, players must use the WASD controls to "strafe" around trees while keeping their mouse aimed at the target.

Figure 8: Level 1 Jungle Hunt

Level 2: The Dungeon Escape

Level 2 shifts the gameplay from exploration to high-intensity survival. This "Escape Dungeon" style level forces the player into a side-scrolling progression within a top-down perspective.

- **The Chasing Hazard:** A lethal, moving **Spike Wall** constantly advances from the left side of the screen. If the player lingers too long or gets stuck on geometry, the spikes result in an instant loss of life.
- **The Pumpkin Threat:** This level introduces the **Pumpkin Enemy**. Unlike the Bats, these enemies act as "stationary turrets" or slow-moving tanks that fire multiple projectiles at once. Players must decide whether to stop and fight the Pumpkin for a high score or use a **Speed Ring** to dash past its "bullet-hell" fire pattern to stay ahead of the spikes.
- **Navigation:** With the top and right paths often blocked by dungeon walls, the player is forced into a narrow corridor, making projectile dodging significantly more difficult.

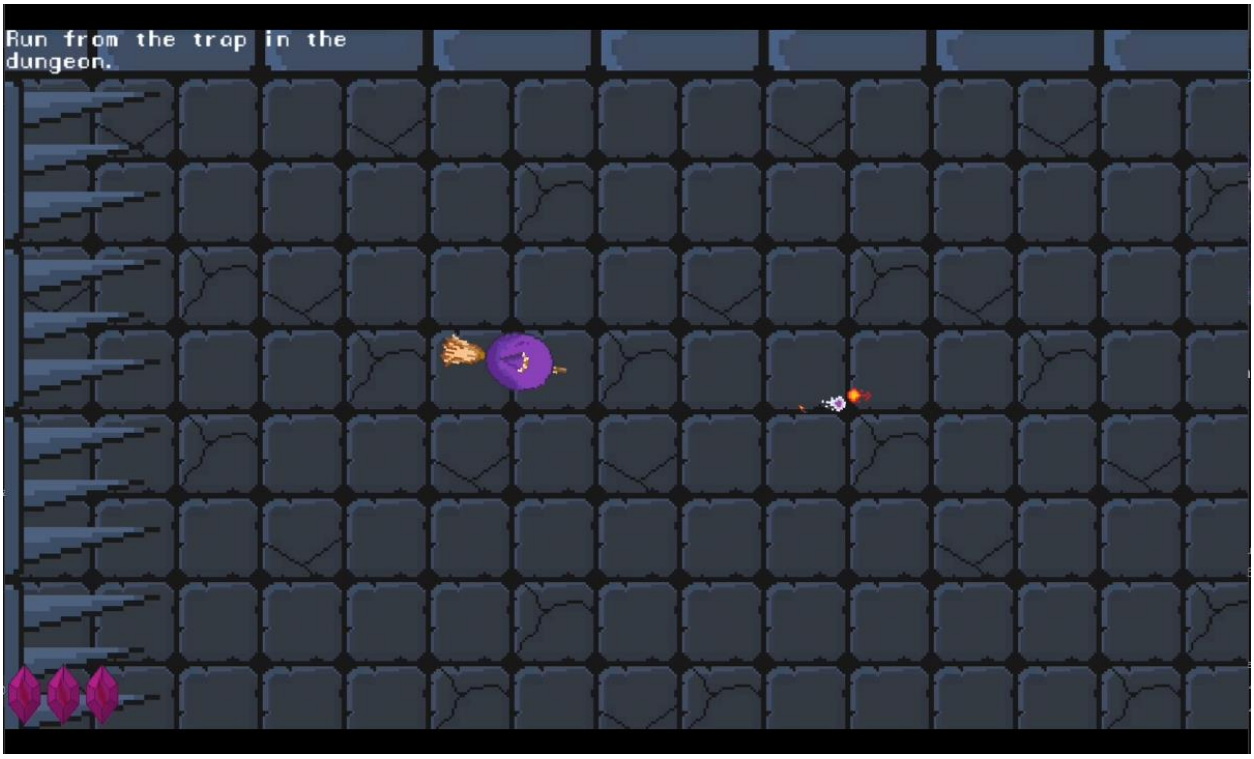

Figure 9: Level 2 Dungeon

Level 3: Survive & Escape

The final level is a "Gauntlet" or "Arena" style challenge that tests everything the player has learned. There is no destination—only time.

- **The Arena:** The player is trapped in a confined square arena surrounded by impenetrable walls. A **60-second countdown** is displayed prominently on the UI.
- **Dynamic Spawning:** Enemies (Bats, Ghosts, and Pumpkins) spawn at an increasing rate as the timer ticks down. By the final 10 seconds, the arena becomes a chaotic "Survival" zone where the player must balance shooting and dodging.
- **Resource Management:** To survive the full minute, players must strategically time their collection of **Health Potions** that spawn in the corners. Picking them up too early might leave the player vulnerable during the final, high-density wave of enemies.
- **The Escape:** Once the timer hits zero, a passage opens in the wall. The player is not safe until they physically reach that exit, adding a final "dash" mechanic to the end of the game.

Figure 10: Level 3 Arena

4 User Interface (UI) & Sound Design

4.1 UI Elements

HUD: Displays the 3-point health system (bottom-left), current score, high score, and a live countdown timer for survival levels.

Menus: A fully functional Main Menu (New Game, Level Select, Instructions, Exit) and an in-game Pause Menu (ESC key).

Feedback: Post-level screens notify the player of success and provide a prompt to progress.

Figure 11: Main menu

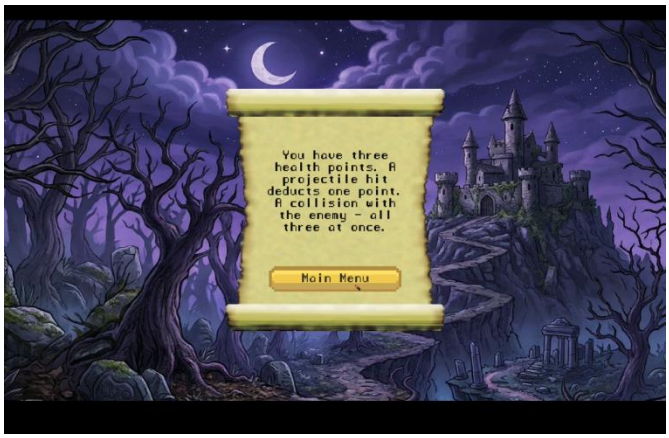

Figure 12: Instructions

Figure 13: Levels

Figure 14: Level complete notice

4.2 Audio Implementation

Atmosphere: Specific lobby music and thematic level tracks are used to build immersion.

Feedback Loops: Unique SFX for enemy damage/death, player "hurt" sounds, and distinct "ping" sounds for picking up potions or passing through rings to provide satisfying auditory cues.

5 Control Guide

Input	**Action**
W, A, S, D	Move Witch (Up, Left, Down, Right)
Mouse Move	Aim Projectiles
Left Click	Fire Projectile
ESC	Pause Game
Collision	Automatically pick up Rings/Potions

Table: 1

6 Creative Feature: Dynamic Combat & Feedback Systems

While many top-down shooters rely on static movement, **Witch Hunt** elevates the experience by integrating a suite of reactive systems that prioritize player precision and visual clarity. Instead of a traditional "walk and shoot" approach, the game utilizes a combination of high-stakes enemy AI and sophisticated feedback loops.

- **Precise Mouse-Controlled Aiming**

The core of the combat is the **Decoupled Aiming System**. By separating the player's movement (WASD) from the aiming direction (Mouse), the game allows for high-level "strafing" maneuvers. The projectile trajectory is calculated in real-time based on the mouse's screen-space coordinates. This implementation is crucial for survival, as it allows the player to retreat from a swarm of Bats while precisely picking them off without turning their back on the danger.

- **Reactive Enemy Archetypes: The Suicide Bats**

A standout element of the game's challenge is the **Suicide Bat AI**. These are not passive enemies; they are programmed with a specific "aggro-radius." Once the player enters their zone, the Bats switch from an idle state to a high-speed pursuit mode. Because they deal massive damage upon collision (potentially ending a run instantly), they force the player to prioritize targets under pressure, creating a tense "risk vs. reward" dynamic during exploration.

- **Dynamic Fading Health Bars**

To maintain a clean, immersive aesthetic without sacrificing vital information, I implemented **Contextual Health Bars** for Ghosts and Pumpkin enemies.

  - **The Mechanic:** Rather than cluttering the screen with permanent UI overlays, health bars only instantiate or become visible when an enemy takes damage.
  - **The Implementation:** Using an Alpha-fading script, the health bar remains visible for a few seconds post-hit to show the player their progress, then smoothly fades away. This gives the game a modern, "polished" feel where the UI only speaks to the player when necessary.

- END OF THE REPORT-

Useful Links

Gameplay Video - Link

Game Files - Link